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Expanding the Dimensions of Epistemic Cognition: Arguments From Philosophy and Psychology

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Psychological and educational researchers have developed a flourishing research program on epistemological dimensions of cognition (epistemic cognition). Contemporary philosophers investigate many epistemological topics that are highly relevant to this program but that have not featured in research on epistemic cognition. We argue that integrating these topics into psychological models of epistemic cognition is likely to improve the explanatory and predictive power of these models. We thus propose and explicate a philosophically grounded framework for epistemic cognition that includes five components: (a) epistemic aims and epistemic value; (b) the structure of knowledge and other epistemic achievements; (c) the sources and justification of knowledge and other epistemic achievements, and the related epistemic stances; (d) epistemic virtues and vices; and (e) reliable and unreliable processes for achieving epistemic aims. We further argue for a fine-grained, context-specific analysis of cognitions within the five components.

In a very active arena of current research, educational and developmental psychologists have investigated human cognitions about epistemic matters. These are cognitions about a network of interrelated topics including knowledge, its sources and justification, belief, evidence, truth, understanding, explanation, and many others. Different researchers have used different terms for these cognitions, including personal epistemology (Hofer & Pintrich, 2002), epistemological beliefs (e.g., Schommer, 1990), epistemic beliefs (Mason, 2003; Muis, 2007; Muis & Franco, 2009), epistemic positions (Perry, 1968/1999), epistemic cognition (Greene, Azevedo, & Torney-Purta, 2008), epistemological reflection (Baxter Magolda, 2004), and reflective judgment (King & Kitchener, 1994). Following Kitchener (2002) and Greene et al. (2008), we adopt the term *epistemic cognition* (EC) as an umbrella term encompassing all kinds of explicit or tacit cognitions related to epistemic or epistemological matters. We believe that EC is worth studying in its own right as a significant domain of human cognition. In addition, EC is important because epistemic beliefs (viewed as beliefs about knowledge and knowing) have been found to predict learning processes and outcomes (e.g., Bråten, Britt, Strømsø, & Rouet, 2011; Hofer, 2001; Muis, 2004).

In a seminal article, Hofer and Pintrich (1997) used both psychological arguments and philosophical considerations to define the scope of personal epistemology. They argued that learners' personal epistemologies consist of systems of beliefs about (a) the nature of knowledge and (b) the processes of knowing. They proposed two dimensions of beliefs about the nature of knowledge: the simplicity versus complexity of knowledge as well as the certainty versus uncertainty of knowledge. They similarly postulated two dimensions of beliefs about the processes of knowing: the sources of knowing (such as authority) and justification (such as the evaluation of knowledge claims and the use of evidence). Many subsequent publications have explicitly drawn on this influential conceptualization of EC.

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Our goal in this article is to extend this landmark framework of Hofer and Pintrich (1997) and to contribute to their project of using philosophical and psychological considerations to clarify the construct of epistemic cognition. However, the conceptualization of EC that we expound differs from the framework of Hofer and Pintrich in two important ways. First, we argue for the inclusion of several new components and subcomponents of EC. Second, we argue for a more specific, fine-grained analysis of cognitions within several of the dimensions that Hofer and Pintrich discussed (including sources and justification).

Building on philosophical scholarship, we thus present an expanded framework for models of EC. In this framework, EC consists of a network of interconnected cognitions that cluster into at least five distinguishable components. We next provide brief initial definitions of each component.

1. *Epistemic aims and epistemic value.* Epistemic aims are a subset of the goals people adopt, specifically those goals related to inquiry and finding things out. Epistemic aims discussed by philosophers include knowledge, understanding, and true beliefs. When people achieve these aims, we refer to the resulting products (such as knowledge, understanding, or true beliefs) as *epistemic achievements or accomplishments.* Epistemic value refers to the worth of particular epistemic achievements. For example, a person who believes that scientific knowledge is worth attaining because it supports economic growth has a belief that scientific knowledge is valuable for practical reasons.
2. *The structure of knowledge and other epistemic achievements.* Knowledge and other epistemic achievements can be viewed as having structure, such as a simple structure or a complex structure. The second component of our framework specifies multiple dimensions of structure, including not only simplicity-complexity but also other dimensions. It also emphasizes very specific epistemic structures (such as people's understanding of microbiological mechanisms) that can play an important role in cognition.
3. *The sources and justification of knowledge and other epistemic achievements, together with related epistemic stances.* The source of knowledge refers to where knowledge originates, such as a person's reasoning or perceptual processes. Justification refers to people's reasons for their beliefs. Epistemic stances refers to the attitudes that people take with respect to an idea, such as believing it, doubting it, tentatively endorsing it, holding it as absolutely certain, or entertaining it as a possibility. The first two of these components (sources and justification) are roughly the same as the dimensions discussed by Hofer and Pintrich (1997), although our conceptualization differs in several important respects. The stances component includes what Hofer and Pintrich described as beliefs about certainty.
4. *Epistemic virtues and vices*. Epistemic virtues are praiseworthy dispositions of character that aid the attainment of epistemic aims, such as intellectual courage and open-mindedness. In contrast, epistemic vices are those dispositions that hinder the achievement of epistemic aims.
5. *Reliable and unreliable processes for achieving epistemic aims*. This component concerns the processes (e.g., cognitive and social processes, inquiry methods) by which knowledge (and other epistemic aims) are achieved. For example, a student might regard extended argumentation with peers as a good process for developing knowledge of history.

In addition to specifying these five components, we argue that any EC framework should also distinguish between individual and social dimensions of EC for each component. For example, a person may have beliefs about reliable individual processes for producing knowledge (e.g., particular cognitive strategies for finding things out) or reliable social processes for producing knowledge (e.g., institutional processes of peer review). However, due to considerations of space, we do not systematically discuss the individual and social aspects of each of the components.

Of the five components, current work in educational psychology has focused principally on Components 2 and 3, specifically the complexity, certainty, sources, and justification of knowledge. There has been some additional EC work addressing one aspect of Component 4, epistemic virtues. On the basis of both philosophical and psychological considerations, we argue for an expanded framework of EC that includes additional Components 1 and 5. The expanded framework also includes a range of new topics within Components 2, 3, and 4 that have not yet been investigated by EC researchers. Finally, the expanded framework emphasizes the fundamental importance of more fine-grained, context-specific cognitions, thus suggesting new avenues for EC research. We argue that epistemic cognitions often vary across situations. Our framework explains some of the respects in which cognitions about sources, justification, and reliable processes are situation specific, and it identifies more specific, finer grained epistemic cognitions with the potential to predict situation-specific learning patterns.

Our aim is to develop a psychological framework of epistemic cognition that addresses the full range of psychologically important epistemic phenomena, concepts, and issues. The development of our framework has been strongly influenced by the work of contemporary epistemologists. In drawing on philosophy for insights into epistemic cognition, we follow the recommendations of other educational psychologists (Greene et al., 2008; Murphy, Alexander, Greene, & Edwards, 2007), who have argued that research on EC can profit from more extensive engagement with philosophy.

We have found philosophical scholarship to be highly relevant to psychological theories of epistemic cognition.

Philosophers working in the field of epistemology have investigated a tapestry of interconnected phenomena, concepts, and issues (call them *topics*), which they judge to be important to understanding epistemic matters. Further, many of these philosophers view themselves as studying epistemic cognition, given that they investigate the nature of human epistemic achievements, as well as the ways in which these achievements are achieved (Bishop & Trout, 2005; Giere, 1988; Goldman, 1986; see Alston, 2005, pp. 1–3). In light of these investigations, we present a range of topics that have been extensively investigated by epistemologists but that have not been discussed much or at all by EC researchers. We argue that cognitions related to each of these topics are likely to be critical for explaining learning processes and outcomes.

The outline of this article is as follows. First, we review existing psychological work on EC, focusing particularly on the scope of the topics considered in most current research. Second, we show that philosophers view the scope of epistemic cognition in broader terms than have most EC researchers. Third, we discuss each of the new components of EC that we propose to include in our expanded framework. We present both philosophical warrants and psychological arguments for including each component in our framework, and we highlight similarities and contrasts with current EC research.

PSYCHOLOGICAL CONCEPTUALIZATIONS OF THE SCOPE OF EPISTEMIC COGNITION

In this section, we examine psychological research on epistemic cognition. We begin with a brief summary of research on developmental change in EC and then focus on research that treats EC as a multidimensional system of beliefs. We focus particularly on the dimensions or components that have been the focus of current research.

Developmental Models of Epistemic Cognition

The earliest psychological research on EC traced trajectories of epistemic development in college students (Perry, 1968/1999). This developmental work has been elaborated and extended by a number of other researchers, including Baxter Magolda (1992), King and Kitchener (1994), and D. Kuhn and Weinstock (2002). Developmental theories characterize epistemic development as a progression through a series of stages. D. Kuhn's approach is typical (D. Kuhn & Weinstock, 2002). In her theory, very young children are initially direct realists, regarding knowledge as a direct copy of reality. This is followed by an "absolutist" stage in which ideas are viewed as definitively right or wrong. Next is a multiplist stage, in which different and conflicting ideas are viewed as equally tenable; all opinions are equally good or valid. Finally, at the evaluativist stage, knowledge is considered as uncertain, but tentative conclusions are possible because claims are judgments to be evaluated "according to criteria of argument and evidence" (D. Kuhn & Weinstock, 2002, p. 125).

Although theories of developmental progressions have not typically been articulated in terms of independent dimensions of beliefs (but see Greene, Torney-Purta, & Azevedo, 2010; Greene et al., 2008, for an exception), many of the epistemic constructs that appear as dimensions in theories of epistemological beliefs (e.g., Hofer & Pintrich, 1997) also figure in developmental theories. For example, D. Kuhn and Weinstock's (2002) model discusses developmental changes in terms of changes in beliefs about the certainty of knowledge and the sources of knowledge, with more sophisticated participants being less likely to view knowledge as certain, less likely to accept an external authority as a source of knowledge, and more likely to accept critical reasoning as a source and justification. These developmental constructs of certainty, sources, and justification are among the constructs that Hofer and Pintrich (1997) posited as fundamental dimensions of epistemological beliefs in their multidimensional framework.

Epistemic Cognition as Multidimensional Systems of Beliefs

One prominent approach to analyzing epistemic cognition treats it as a multidimensional system of independent beliefs. Much current EC research is grounded in this approach, and our revised framework bears some resemblance to these multidimensional models.

Schommer's (1990) groundbreaking work in this area specified the following dimensions of epistemological beliefs: the extent to which students believe that (a) knowledge is complex, (b) knowledge is certain, (c) learning is quick, (d) the ability to learn is innate, and (e) knowledge is handed down from authority rather than derived from reason. Schommer (1990) developed a widely used questionnaire for assessing these beliefs; however, across many studies using this questionnaire, factor analysis has often not yielded this exact set of five factors (Buehl, 2008). Other researchers have developed measures that build on Schommer's framework. For example, Jehng, Johnson, and Anderson (1993) developed a scale that included four of Schommer's dimensions but replaced her subscale on the complexity of knowledge with a subscale containing items on whether learning is best approached in an orderly fashion (see also Kardash & Scholes, 1996; Schraw, Dunkle, & Bendixen, 1995).

Hofer and Pintrich (1997) began their influential article with a definition of epistemology as "an area of philosophy concerned with the nature and justification of human knowledge" (p. 88). On this basis, they argued that Schommer's EC dimensions involving learning (quick learning and innate ability) are not epistemic because these beliefs are not centered on the nature and justification of knowledge, and they proposed instead the four dimensions discussed earlier. Hofer (2000) developed a questionnaire focused on the dimensions identified by Hofer and Pintrich (1997). In a factor analysis, the dimensions of certainty and simplicity merged into a single factor. There was a justification factor (personal justification) and a source factor (authority as a source). A fourth, unplanned factor also emerged—the extent to which scientists were seen as able to attain truth.

Since the publication of Hofer and Pintrich's (1997) four-dimension model of epistemological beliefs, many researchers have explicitly drawn on this framework (e.g., Greene et al., 2008; Mason & Scirica, 2006; Muis, Bendixen, & Haerle, 2006; Muis & Franco, 2009; Qian & Alvermann, 2009; Stathopoulou & Vosniadou, 2007; Tsai, 2008; Weinstock & Cronin, 2003). Some EC researchers have also continued to use the Schommer (1990) framework (e.g., Bråten & Strømsø, 2006; K.-W. Chan & Elliott, 2002). We note, however, that there is considerable overlap between the two approaches, with agreement on the dimensions of certainty, complexity, and justification (with authority as justification or as source). The primary difference concerns whether EC models include beliefs about quick learning and innate learning.

In addition to Hofer and Pintrich (1997), a number of EC researchers have based their framing of the scope of EC on a definition of the scope of the discipline of epistemology. For example, Greene et al. (2008) emphasized the importance of cognitions related to justification in models of EC, given the centrality of justification to epistemology. Muis (2008) wrote, "Consistent with philosophical notions of epistemology, this model focuses on general beliefs about how knowledge is derived and how knowledge is justified" (p. 182). Schraw and Olafson (2008) viewed epistemology as "the study of knowledge and knowing" (p. 25). Similar definitions recur in EC research (e.g., Gottlieb, 2007; Kardash & Scholes, 1996; Murphy et al., 2007; Tsai, 2008).

Researchers investigating epistemological beliefs have studied the degree to which these beliefs are domain-general or domain-specific (e.g., Buehl & Alexander, 2005). Domain-general beliefs are robust across multiple domains or disciplines (e.g., a student might believe that beliefs in mathematics, science and history can be held with certainty). Domain-specific beliefs vary from domain to domain (e.g., a student believes that certainty is achievable in mathematics but not in history). Beliefs may also vary by topic within a single discipline (Elby & Hammer, 2001). For example, a student might believe, within the single domain of biology, that scientists' claims about cell organelles are certain, whereas evolutionary claims are highly uncertain. In an extensive review, Muis et al. (2006) concluded that epistemic beliefs have both domain-general and domain-specific aspects.

DEMONSTRATING THE PSYCHOLOGICAL USEFULNESS OF POSITED DIMENSIONS OF EPISTEMIC COGNITION

EC researchers demonstrate the psychological utility of the components in their frameworks of EC in several ways (Hofer, 2001). One approach, employed by EC researchers who work with multidimensional belief frameworks, is to develop measures of students' beliefs and then to use these measures to explain and predict learning processes and outcomes. Many studies have linked dimensions of epistemic beliefs to learning processes and outcomes on a variety of tasks. For example, Kardash and Scholes (1996) reported that students who believed that knowledge is certain were less likely to write conclusions regarding a controversial topic that took into account the partly conflicting evidence they had read. In a recent review, Bråten et al. (2011) summarized many reported correlations between (a) learners' epistemic beliefs about the certainty, complexity, sources, and justification of knowing, and (b) the learning processes and outcomes as students learned from multiple documents. Similarly, Muis (2007) summarized evidence that epistemic beliefs are associated with a variety of self-regulated learning processes, including the strategies and evaluative standards that learners use (for earlier reviews, see Hofer, 2001; Mason, 2003). The significant correlations of EC components with learning processes and outcomes are viewed as providing evidentiary support for the psychological utility of the hypothesized EC components.

Despite the many reported statistically significant correlations between epistemic beliefs and learning processes and outcomes, Schraw and Olafson (2008) noted a problem of "low predictive validity between epistemological factors used in ongoing research and various outcome variables" (p. 29). The correlation coefficients and the proportion of variance in learning processes and outcomes that is explained by epistemic belief factors are often relatively low. These low correlations suggest either that epistemic beliefs do not strongly influence learning or that the ways in which epistemic beliefs have thus far been conceptualized do not yet adequately model the epistemic cognitions operative in learning processes.

Further, there is evidence of problems with the validity of measures of beliefs. In a study of three frequently used measures of epistemic beliefs, DeBacker, Crowson, Beesley, Thoma, and Nita (2008) concluded that there are substantial psychometric problems with these instruments, suggesting that there are problems with existing conceptualizations of the domain of personal epistemology. The low predictive validity and intercorrelations of existing EC measures may stem in part from problems with the operationalization of epistemic constructs in the existing assessment instruments. However, a second reason may be that current instruments simply do not measure the most important aspects of EC. There is thus good reason to explore new directions in conceptualizing EC, in the hope of obtaining higher predictive validity between measures of EC and measures of learning processes and outcomes. We argue that measuring EC in terms of the components we propose has potential for better prediction in psychology. One reason is that the additional components in our extended framework may include cognitions that prove particularly important for understanding processes of learning and reasoning. A second reason is that a more fine-grained analysis of students' epistemic cognitions—one more finely tuned to particular learning situations—can enable better explanation and prediction of learning processes and outcomes.

DEFINING THE SCOPE OF EPISTEMOLOGY: A BRIEF OVERVIEW OF PHILOSOPHICAL WORK

One objection that might be raised against our proposal for an expanded framework of EC is that some of the definitions of epistemology offered by philosophers, and used by EC researchers to conceptualize the domain, do not seem to fit our expanded conceptualization. For example, as Muis et al. (2006) noted,

> The Cambridge Dictionary of Philosophy (Audi, 1999) defines epistemology as 'the study of the nature of knowledge and justification: specifically, the study of (a) the defining features, (b) the substantive conditions or sources, and (c) the limits of knowledge and justification' (p. 273). (p. 6)

This definition seems congruent with the idea that EC encompasses the "sources" and "justification" of knowledge; it makes no explicit mention of the new topics that we propose.

In reality, however, epistemologists view their field much more broadly than this definition might seem to suggest. The definition quoted by Muis et al. (2006) was written by Paul Moser, who authored the dictionary's entry on epistemology. Three years later, Moser (2002a) provided a similar definition of epistemology as "the study of the nature of knowledge and justification" (p. 3) in the introduction to his edited volume, *The Oxford Handbook of Epistemology* (Moser, 2002b). However, Moser's (2002b) handbook includes discussions addressing epistemic aims, epistemic virtues and vices, reliable and unreliable processes for producing knowledge, and social aspects of epistemology, and Moser's (2002a) introduction to the handbook addresses each of these topics. Moser clearly interpreted the *study of the nature of knowledge and justification* quite broadly, encompassing epistemological topics that correspond to each of the five components of EC that we propose.

Other prominent epistemologists have explicitly characterized epistemology more broadly. For example, Alston (2005) wrote that epistemologists investigate "the operation and condition of our cognitive faculties—perception, reasoning, belief formation, the products thereof—beliefs, arguments, theories, explanation, knowledge" (pp. 2–3). In discussing the scope of epistemology, other philosophers have explicitly emphasized topics that correspond to our five components, including epistemic aims (e.g., Bishop & Trout, 2005), epistemic virtues (e.g., Sosa, 2007), and reliable processes for achieving epistemic aims (e.g., Goldman, 1986). These philosophers have considered dimensions of epistemology that go well beyond the dimensions of simplicity, certainty, sources, and justification.

Although definitions of the kind just described are suggestive, discerning the true scope of epistemology requires examining the actual practice of epistemologists, with a focus on what they actively investigate. To determine this, we carried out an extensive review of the literature, guided by the second author, a philosopher. We examined 10 handbooks and anthologies of epistemology published in the last two decades and the contents of five leading journals in epistemology from 2000 to 2009. We compiled a list of more than 100 widely cited books in contemporary epistemology; at least one (and usually two) of us read each of these books.[1] The goal of our review was to identify epistemological topics that are deemed by epistemologists to be important for understanding epistemic cognition but that have not yet featured in psychological research. We sought in particular to identify topics with the potential to fruitfully enrich models of EC. This article is organized around a set of core epistemological topics that were identified in this search.

Many of the new topics identified for EC research have emerged as flourishing topics of epistemological investigation only within the last several decades. Why is it that these new topics have arisen in contemporary epistemology? Several historical shifts have contributed to these new developments (see Alston, 2005). One particularly important shift is the shift away from a predominant concern with the problem of radical skepticism—the philosophical stance that human knowledge is unattainable (Kvanvig, 2003). When focused on radical skepticism, epistemological debates have focused on whether humans can have knowledge of even their most basic perceptual beliefs, such as whether they can be confident that they are not dreaming or otherwise deceived. More recently, however, as Alston (2005) wrote, "the exclusive attention to these matters [radical skeptical doubts] has been challenged recently from a variety of directions" (p. 3). One prominent challenge has been the rise of *epistemological naturalism*, which examines the natural and causal processes by which humans form and evaluate their beliefs (Goldman, 1986, 1999; Kitcher, 1993; Kornblith, 1985). Rather than examining whether we can have knowledge at all, epistemological naturalists have examined how individuals and communities (such as scientists and scientific communities) generate knowledge. Crucially, these new directions involve investigation of the more complex kinds of beliefs and belief-forming practices (such as scientific and historical beliefs) that are of interest to educational psychologists, rather than the simple forms of knowledge that were the focus of debates about radical skepticism. More generally, the new directions taken by epistemologists have included attention to epistemic

[1] A list of these sources can obtained from the first author.

aims other than knowledge and true belief (aims such as understanding), the development of new epistemological theories such as virtue epistemology and reliabilism (the basis of the fourth and fifth components of our proposed framework), as well as new theories of social dimensions of epistemology.

We believe that our focus on contemporary epistemology explains some of the differences between our EC framework and the frameworks developed by other researchers. We believe further that the many topics explored by contemporary epistemologists are generally more relevant to EC research than is the narrower range of topics that dominated much of traditional epistemology. Our EC framework includes many relevant topics discussed by philosophers; however, we readily acknowledge that there are many other topics of potential interest that we lack space to address.[2]

THE FIVE COMPONENTS OF EPISTEMIC COGNITION IN THE EXPANDED FRAMEWORK

We turn now to the main section of the article, in which we discuss each of the five components of our framework. In discussing each component, we explain the topics that make up that component and contrast these topics with those currently investigated by EC researchers. We note philosophical and psychological evidence that people do in fact exhibit the cognitions that we propose. We further argue that the framework is likely to be psychologically fruitful in predicting and explaining human learning and reasoning.

There are two reasons why we expect greater predictive and explanatory power for the components of our framework. First, our proposed framework expands the range of topics investigated, and some of these new topics may be important in predicting and explaining learning and reasoning processes. Second, our framework indicates that EC is often highly specific and often varies from situation to situation. To predict and explain learning in a given situation, one needs to know the specific epistemic cognitions that are operative in that situation. Our proposed framework helps identify these specific cognitions to afford more fine-grained, situational explanations of learning processes.

Our discussion of each component includes discussions of beliefs as part of epistemic cognition and are thus directly relevant to research that investigates students' epistemic beliefs through interviews or questionnaires. However, we want to emphasize that our framework does not presuppose that people have the ability to articulate explicit epistemic beliefs (as when answering questions in an interview) or that they can meaningfully express beliefs in response to abstract Likert items such as "In history, the truth means different things to different people" (from Greene et al., 2010). On the contrary, people's epistemic beliefs may be *tacit* rather than *explicit* (Hofer & Pintrich, 1997; Schraw & Olafson, 2008). A tacit belief is a belief that people may be unable to verbally express but that could potentially be inferred from their actions. For example, a science student who regularly justifies knowledge claims by appealing to her personal experiences could be characterized as having a tacit belief in justification by personal experience, even if she cannot articulate such a belief in the abstract, and even if she does not agree in the abstract to the statement, "I prefer to believe things that I can base on my personal experience." The unarticulated belief can nonetheless be inferred from regularities in her practices.

What we have called *tacit epistemic beliefs* might better be called *epistemic commitments* (Chinn & Brewer, 1993). Some theorists may be uncomfortable with the idea that one can have a tacit "belief" that cannot be expressed, and the term *epistemic commitment* avoids reference to such beliefs. An epistemic commitment reflects a tendency to act in specified ways, such as a proclivity to provide justifications based on personal experience. In this article, we use the terms *commitments* and *tacit beliefs* interchangeably. Whenever we use the term *beliefs*, we consider both beliefs that may be explicitly verbalizable and tacit beliefs or commitments that cannot be accurately verbalized. Although our own theoretical stance is that people have many epistemic commitments that they cannot explicitly verbalize, we are adopting, for the purpose of this article, a neutral stance allowing beliefs to be either explicit or tacit. Thus, we intend our framework to be useful to scholars who work within the epistemic beliefs paradigm as well as to scholars who reject the idea that people have explicit epistemic beliefs.

Epistemic Aims and Value

The first proposed component of the extended framework of EC comprises a cluster of interrelated cognitions related to epistemic aims and epistemic value. Current frameworks for EC, such as Hofer and Pintrich's (1997) framework, do not explicitly include epistemic aims and value. In this section, we explicate this component and argue that these cognitions should be included in any framework for EC.

Epistemic Aims

Epistemic aims are goals related to finding things out, understanding them, and forming beliefs. In this section, we discuss several types of epistemic aims, explain why we believe that they are central to EC, and discuss the context specificity of epistemic aims.

Types of epistemic aims. Philosophers have discussed a variety of epistemic aims. The most extensively discussed aim is knowledge. On the standard philosophical analysis,

[2]Although there is considerable disagreement among many of the epistemologists from whose work we draw, discussion of their debates is beyond the scope of this article.

a person seeking knowledge aims to acquire true, justified beliefs—beliefs that accurately represent particular aspects of the world (at least approximately) and that are supported by adequate reasons.

One central epistemic aim that people might adopt is thus the aim of acquiring *true beliefs*, or at least beliefs that approximate or approach the truth (Niiniluoto, 2002). In addition, epistemologists have discussed the epistemic aim of avoiding false beliefs. However, an individual who aims to avoid false beliefs could succeed simply by refusing to adopt any beliefs at all, which would preclude him or her from acquiring any further knowledge. Hence, philosophers have explored the more complex epistemic aim of achieving a store of beliefs with a high truth to falsity ratio (Goldman, 1986). Crucially, people may vary on their preferred ratio of true to false beliefs. EC researchers might thus productively distinguish between *conservative* believers, who are cautious about adopting new beliefs so as to reduce the possibility of error, and *liberal* believers, who, given their primary aim of increasing their stock of true beliefs, are far more open to adopting new beliefs even at the risk of acquiring more false beliefs. A conservative believer might approach a learning task such as learning educational psychology quite differently from a liberal believer, setting very high standards for adopting new beliefs and thus ending up with a much smaller stock of beliefs about effective learning and teaching practices.

Another distinctive epistemic aim that people might adopt is minimally justified belief. A person who preferred the aim of minimally justified belief over the aim of truth might accept a belief that has some justification even if that justification is particularly weak. Such a person may be uninterested in weighing different justifications against each other to find the truth, because merely having a justification is viewed as sufficient for belief. Debate and argumentation is likely to fail to change such an individual's mind. In contrast, a person aiming for justified true beliefs would be more likely to engage in such debates if they consider them as a means to acquiring truth.[3]

Kvanvig (2003) discussed understanding as an important and distinctive epistemic aim. For Kvanvig, understanding is distinct from knowledge because, although knowledge might consist of a collection of disconnected facts, understanding involves grasping "explanatory connections between items of information," (Kvanvig, 2003, p. 193) and seeing how information fits together.[4] Similarly, some philosophers have argued that the construction of explanations is a central epistemic aim (Kitcher, 1993). Students who aim for understanding or explanation will likely approach learning tasks quite differently from students who aim simply to collect a list of true beliefs (Muis, 2007; Rosenberg, Hammer, & Phelan, 2006). For example, a student who aims for explanations in history class may not be satisfied just to learn the tragic sequence of events of the Great Depression; the student may seek deeper economic explanations of why it began and why it lasted so long. In general, people can adopt a variety of epistemic aims in different learning situations, and the aims they adopt have the potential to influence the learning processes in which they engage.

The centrality of epistemic aims. For two reasons, we argue that epistemic aims are an essential component of EC. The first reason is conceptual. In our expanded framework, epistemic cognitions are cognitions directed at epistemic aims and their achievement; epistemic aims are central to EC because aims determine whether other cognitions should be classed as epistemic or not. Many beliefs can be ruled out as nonepistemic because they are not directed at epistemic aims. For example, a student's belief that "I like class work best when it really makes me think" (Midgley et al., 2000) is not epistemic because no epistemic aim is invoked; mere thinking is not an epistemic aim such as knowledge or understanding. A belief that "I consider argument an exciting intellectual challenge" (Infante & Rancer, 1982) is not an epistemic belief because people could agree to this statement despite having no desire to pursue any epistemic aim (e.g., truth or knowledge); rather, they might simply enjoy the challenge of arguments in order to demonstrate their argumentative virtuosity and to win the argument, regardless of whether they attained truth, knowledge, or understanding. In contrast, a belief that engaging in argumentation with others helps one develop more coherent knowledge is an epistemic belief because it is about a process of achieving a certain kind of knowledge structure. Epistemic aims are the end to which all other epistemic beliefs and activities are directed; it would thus be a conceptual error to exclude epistemic aims from frameworks of EC.

The second reason for including epistemic aims as part of EC is to enhance the predictive and explanatory power of EC models. It is impossible to adequately explain or predict learning and reasoning processes without knowing whether people have adopted epistemic aims or which aims they have adopted. To predict people's learning and reasoning processes, it is not sufficient to know only their epistemic beliefs, such as their beliefs about the structure of knowledge, without also knowing the epistemic aims they adopt. For example, consider three students taking a history class. All three believe that historical knowledge is complex. But a belief that historical knowledge is complex does not entail that these students will adopt the aim of constructing such knowledge and attempting to understand it. The first student simply has no intention of trying to develop such an understanding, as

[3]There is indirect evidence from a study by Kruglanski, Peri, and Zakai (1991) that individuals who exhibit a need for closure are pursuing the aim of minimally justified knowledge as we have described it. We discuss need for closure further under epistemic virtues and vices.

[4]The aim of understanding may seem at first appear to be equivalent to the aim of constructing complex interconnected knowledge. However, this is mistaken. Interconnected ideas can be either simple or complex, and one can aim to understand relatively simple as well as highly complex material. Similarly, knowledge can be interconnected without being highly complex.

she would prefer to spend her time on other courses that she finds more interesting. She instead aims for knowledge of a list of a few key points to prepare for her exams. The second student's beliefs about the complexity of historical knowledge actually dissuade her from aiming to construct such knowledge; she is keenly aware of how much work it would take to master the complexity, and, as a consequence, she aims for a more limited understanding. The third student is the only student who both believes that historical knowledge is complex and aims to understand it, and so sets out to master the complexity of the material. This example illustrates the importance of considering learners' epistemic aims as well as their other epistemic beliefs in order to predict their learning processes. Students do not necessarily adopt epistemic aims that are congruent with their other epistemic beliefs; indeed, when they believe that knowledge is complex, this belief may in fact discourage them from setting the challenging aim of attaining complex knowledge. Similarly, if students believe that developing justified beliefs requires a great deal of effort, they may opt instead simply to accept unjustified beliefs.

EC researchers have typically measured epistemic beliefs (such as beliefs about the complexity of knowledge) without explicitly asking whether students set aims corresponding with these beliefs. There is some evidence that mismatches between epistemic aims and other epistemic beliefs exist and may even be common. Hammer (1989) presented an example of a physics student who was aware that physics knowledge was meaningful and interconnected but decided that the fast pace of the course left too little time to allow for the aim of understanding those concepts; instead, she opted for more rote memorization approaches to get through the course. Elby (1999) reported that there were many such students in university physics classes. Low correlations between measures of epistemic beliefs and learning processes and outcomes may arise because many students may fail to adopt epistemic aims that are commensurate with their beliefs about the certainty, structure, and justification of knowledge. We argue that it is thus critical to examine epistemic aims alongside other beliefs in order to understand learning processes.[5]

Context specificity of epistemic aims. Philosophers note that people adopt different aims in different situations (Bishop & Trout, 2005; Morton, 2006). In one class period, a student may adopt the aim of developing a good scientific explanation of photosynthesis. The next day the same student may adopt a simpler aim of simply learning a list of facts about plant growth. Students may aim to garner justifications for their beliefs on some occasions but not others. Rosenberg et al. (2006) presented a case study of a science lesson in which a teacher comment prompted a group of students to shift abruptly from an aim of constructing a list of facts to the aim of constructing a coherent explanation. Given that epistemic aims can vary by context, and that the particular epistemic aims an individual pursues might shift over the course of a single conversation, it should be an important goal of EC to identify students' aims in particular situations, in addition to identifying factors that influence which aims will tend to predominate in various situations. To understand learning and reasoning in particular situations, researchers must understand which aims students adopt in those situations as well as why they adopt those aims. Both situation-specific questionnaires and observations of students' discourse could provide information about the aims that they adopt in particular settings.

Epistemic Value

People adopt various epistemic aims because the resulting epistemic achievements have value to them. Philosophers throughout history have asked which epistemic achievements are most valuable and why these achievements have value. For example, Plato argued that knowledge is more valuable than true belief because it is justified, and justified beliefs are less likely than unjustified beliefs to be true due to luck (e.g., a lucky guess). Kvanvig (2003) contended that understanding (which involves grasping explanatory connections) has more value than either knowledge or true belief (which may simply comprise a list of unrelated propositions).

Baird (2004) and Staley (2004) have observed that some general types of knowledge in science are treated (perhaps wrongly) as more valuable than others. In particular, theoretical knowledge is often viewed as more valuable than practical knowledge of how to make things (such as equipment), although both kinds of knowledge are indispensible in science (research could not be conducted without specially designed apparatus). However, a recent study of the epistemic beliefs of research chemists showed that they do value "good hands," the ability to do good bench work in the laboratory (Samarapungavan, Westby, & Bodner, 2006). People might thus differ in the relative value they place on different kinds of epistemic achievement.

Philosophers have observed that some knowledge appears to be more valuable than other knowledge. For example, few people would say that it is valuable to know how many pages there are in each book in their university library. In contrast, most people would probably agree that knowledge of how to cure cancer would be extremely significant and valuable.

[5]Our discussion of the aim of understanding raises a question of whether motivational researchers' measures of mastery goals (characterized as goals to learn, understand, and master material) capture the epistemic aim of understanding. In fact, some items in some measures do assess whether students value the aim of understanding, such as "It's important to me that I thoroughly understand my class work" (Midgley et al., 2000). However, many items in these measures do not measure the adoption of epistemic aims, such as "The opportunity to do challenging work is important to me" (see Hulleman, Shrager, Bodmann, & Harackiewicz, 2010, for further examples). This item measures the value placed on challenging work but does not directly assess whether the student values or aims to acquire truth, justified beliefs, or other epistemic aims; challenging work may be valued simply because it is fun. To properly assess epistemic aims, more carefully targeted measures are needed.

Many people also seem to view knowledge of the origins of the universe and of humans to be particularly valuable. Some truths seem less valuable or significant than others because they seem incomplete; for example, scientists could truthfully state in the late 1800s that all planets except Mercury follow an ellipse, but scientists were dissatisfied with this true (though incomplete) description. They sought instead for an account that could explain the motion of all of the planets, including Mercury (Elgin, 2006), and viewed this unified explanation as more valuable than the less unified, descriptive account. From considerations such as these, many philosophers have concluded that people particularly value *significant truths* and have developed various accounts of epistemic significance (Goldman, 2002; Kitcher, 2001; Zagzebski, 1996). This raises important but as yet unanswered questions about what makes different truths significant for different people.

To date, EC researchers have not explicitly incorporated beliefs about epistemic value in their EC frameworks. We suggest two lines of possible investigation. First, EC researchers could examine people's beliefs (explicit or tacit) about the value and significance of different epistemic achievements. For example, researchers could investigate the extent to which people value knowing how to do things, as opposed to acquiring knowledge of theories. Using interview protocols, researchers could provide students with many different kinds of knowledge and ask them to explain the relative value of each. Some students might stress the significance of knowledge that addresses important societal problems. Other students might see only personally relevant knowledge as significant. Still other students might exhibit a preference for broad, powerful ideas that attempt to explain many phenomena, even if imperfectly, whereas others value descriptive, nonexplanatory knowledge as highly as powerful explanations. Through this means, researchers might develop profiles of the kinds of epistemic achievements that are valuable to different students, as well as why the students value them.[6]

EC researchers could also investigate how students' epistemic values and their beliefs about what is epistemically valuable are associated with their learning processes and outcomes. Other things being equal, we expect that people will be more likely to pursue epistemic achievements that they deem to be valuable or significant. Teachers who value content knowledge more than inquiry skills may resist inquiry teaching. Students who value declarative theoretical knowledge more than practical, hands-on knowledge may devote little time to mastering the lab techniques needed to conduct chemistry experiments. Samarapungavan et al. (2006) found that some graduate students in chemistry fail to achieve full success due to a lack of hands-on laboratory skills; using our framework, we would hypothesize that these students may not have valued hands-on knowledge as much as theoretical knowledge of chemistry. Thus, beliefs about the value of different kinds of knowledge could play an important role in the trajectories by which expertise develops.

Whether learners adopt aims in line with their judgments of value is likely to depend on their judgments of the costs of pursuing the aims relative to the value of the resulting achievements (Bishop & Trout, 2005). Students who value particular knowledge will seek to acquire it only if they judge that the value of the knowledge exceeds the costs of acquisition (e.g., the time and effort required). Thus, an important task for EC research should be to examine students' relative assessments of the value and costs of achieving different epistemic aims (cf. Greene & Azevedo, 2007).

Conclusions

In summary, we contend that epistemic aims and value should be included as an indispensible component of frameworks of EC. Epistemic value is important because people weigh epistemic value against costs in deciding which aims to adopt. Aims are important because the epistemic aims that people adopt (if any) influence how they approach learning tasks. We have argued that relatively low correlations between epistemic beliefs (such as beliefs about the complexity of knowledge) and learning and reasoning processes may arise in part from not taking into account whether people adopt epistemic aims in particular situations and, if so, which aims they adopt.

Structure of Knowledge and Other Epistemic Achievements

The second component in our framework for EC is the structure of knowledge and other epistemic achievements. This component is roughly comparable to the "structure of knowledge" dimension advanced by Schommer (1990) and by Hofer and Pintrich (1997). Our framework, however, includes the structure of other epistemic achievements such as explanation.

Most EC researchers have treated people's beliefs about the structure of knowledge as a single dimension involving a belief that knowledge is simple at one pole and complex at the other (see Buehl, 2008, for a review). For example, Schommer-Aikins and Easter (2008) characterized this belief along a single continuum ranging from "knowledge is organized as isolated facts" to "knowledge is structured in integrated, complex, and sometime [*sic*] ambiguous concepts" (p. 923). This dimension of belief has frequently been found to predict students' learning (e.g., Mason, 2003; Muis, 2004). For example, beliefs about simple versus complex knowledge have been used to predict the performance of students learning statistics (Schommer, 1992).

[6]Measures of task importance by expectancy-value theorists have some similarities to epistemic value. However, typical items in these measures are very general, such as rating how important it is to do well in math (Wigfield & Eccles, 2000). We advocate a more detailed profile of the specific kinds of knowledge that students find valuable, as well as why they think these kinds of knowledge have value.

In this section, drawing on philosophical analyses, we outline a more multifaceted view of the structure of knowledge. Our framework differs from the current prevalent conceptionalization of the structure of knowledge in two ways: (a) We view the structure of knowledge as multidimensional rather than unidimensional, and (b) in addition to broad structural dimensions such as simplicity-complexity, we emphasize the importance of more specific structural forms such as mechanisms and causal frameworks.

A preliminary question that arises about the "structure of knowledge" component is whether it should even be considered an *epistemic* dimension. Murphy et al. (2007) and Greene et al. (2008) have argued that students' beliefs about the structure of knowledge should be viewed not as epistemological beliefs about knowledge but instead as ontological beliefs about the structure of the world. We agree that claims about the structure of knowledge might reflect underlying *ontic* commitments (i.e., about the structure of the world), rather than merely *epistemic* commitments about the structure of knowledge. However, whether the structure of knowledge should be viewed as ontic or epistemic crucially depends on whether the knower is a philosophical realist, and—if so—on what kind of a realist she is. A naïve epistemological realist is an individual who believes that there is a relatively straightforward one-to-one correspondence between his or her mental representations (i.e., beliefs) and reality. For this individual, a conceptualization of knowledge as simple might derive from an ontological view that the external world is similarly simple, given his commitment to a one-to-one matching between mind and world. However, for thinkers who do not adopt a naïve correspondence view of knowledge–world relations, there can be substantial dissociations between their view of the (epistemic) structure of knowledge and of the (ontological) structure of reality. For example, scientists may recognize that simplified assumptions allow them to model important aspects of the world, even though those aspects of the world are much more complex than their relatively simple models (e.g., Sober, 1988). On this view, the structure of an individual's knowledge does not have straightforward implications for his or her view of the structure of the world. Elliot Sober, a noted philosopher of biology, has argued that most philosophers treat the claim that knowledge is simple as an epistemic claim about knowledge and not as a claim about the structure of the world. "It is hypotheses, not nature itself, that now are said to be simple" (Sober, 1988, p. 43).

In short, beliefs about the structure of knowledge are directly linked to ontological beliefs only for a naïve realist. For everyone else, such beliefs are at least partly epistemic. Nonetheless, it is likely that epistemic and ontological beliefs are often interrelated. We conclude, therefore, that beliefs about the structure of knowledge are a component of *epistemic* cognition, intertwined with *ontic* cognition.

The Structure of Knowledge as Multidimensional

Philosophical work suggests that the structure of knowledge is a multidimensional space rather than a single dimension of simplicity-complexity. Philosophers have discussed a number of dimensions that constitute this space in addition to simplicity-complexity. These other dimensions may prove to be predictive of students' learning processes. We discuss two possible dimensions next. Others are discussed by Code (1991).

Universality versus particularity of knowledge. Feminist philosophers have argued that scientists have historically viewed knowledge as universal (Code, 1991; Duran, 1991). For instance, physicists have promulgated universal laws that are held to apply at all times and places. Psychologists have sought principles of human psychology that apply to all humans. In contrast, feminist philosophers have argued that knowledge should be viewed as structured very differently—as highly particular, specific, and context sensitive. This might be seen to imply, for instance, that knowledge of human psychology cannot be reduced to a set of universal principles, given the uniqueness of the context of each human being.

We expect that people vary in the extent to which they believe that knowledge is universal or contextual (see also Goldman, 2002). Such beliefs (explicit or tacit) may predict how people approach learning tasks. For example, a belief that knowledge of the social world is general and universal may lead a student who reads a highly detailed case study of the effects of welfare reform to ignore much of the case study's detail and instead to try to abstract one or two general principles. In contrast, a student who believes that knowledge tends to be particularized and contextual may be more likely to attend to and recall the details of the case and try to understand contextualized causal interactions.

Deterministic versus stochastic knowledge. A second dimension along which beliefs about knowledge may vary is the extent to which knowledge is conceptualized as deterministic or stochastic (Salmon, 1989). In most contemporary theories in the natural and social sciences, the world is held to be at least partly probabilistic and therefore not fully predictable. For example, randomness exists at fundamental levels in physical theories. Learners may vary greatly in their beliefs about how stochastic knowledge is. Some students may believe that knowledge is fully deterministic (Grotzer, 2003); these students may have difficulty comprehending texts that present theories in which the world is described as relatively unpredictable.

Thus, philosophical scholarship suggests that there are additional dimensions of the structure of knowledge besides simplicity-complexity. By simultaneously measuring multiple dimensions of beliefs about the structure of knowledge, EC researchers can investigate which of these dimensions best predict learning on a variety of tasks.

The structure of explanation. In addition to examining the structure of knowledge, philosophers have examined the structure of other epistemic achievements such as explanations (Salmon, 1989). One philosophical view is that explanation shows how a phenomenon to be explained is an instance of a general law. On this view, one can explain the growth of a corn plant by noting that most corn plants that have good soil, adequate water, and sunlight will grow. Another view is that explanations provide causal accounts showing what causes the phenomenon; on this view, to explain the growth of a corn plant, one must provide an account of the specific cellular processes that produce plant growth. Brewer, Chinn, and Samarapungavan (1998) observed that very little is known about how children or lay adults conceive of explanations.

More Specific Structural Forms

Although philosophers have sometimes discussed broad aspects of the structure of knowledge (simplicity-complexity, universality-particularity, etc.), they have devoted much more attention to examining knowledge structures at a more specific, fine-grained level. For example, instead of discussing the structure of scientific knowledge in general, philosophers have focused on issues such as the structure of mechanisms in molecular biology (Machamer, Darden, & Craver, 2000), the structure of models in mechanics (Giere, 1988), and specific forms of causal knowledge (Cartwright, 2004; Woodward, 2003).

Machamer et al. (2000) developed a detailed analysis of the structure of mechanisms in molecular biology and neurobiology. They argued that much scientific knowledge in these fields is structured in the form of mechanisms that include entities and their properties, setup conditions, termination conditions, and intermediate activities that the entities are involved in to produce the termination conditions. For Machamer et al., understanding this epistemic form is central to understanding what molecular biologists do and how they think. During inquiry, molecular biologists set out specifically to propose new *mechanisms* with the structure just outlined; similarly, molecular biologists evaluate proposed mechanisms according to whether the various elements are present and adequately supported by evidence. In this manner, knowledge of structures can affect both aims and the justificatory standards used to evaluate the results of inquiry (cf. Muis, 2007).

Recent research by Russ and her colleagues (Russ, Coffey, Hammer, & Hutchison, 2009; Russ, Scherr, Hammer, & Mikeska, 2008) has shown that even first-grade students grasp some components of the mechanisms discussed by Machamer et al. (2000). There is thus evidence that these knowledge structures are psychologically meaningful, even for young children.

We propose that learners' beliefs about more specific structures such as the mechanisms of molecular biology will often prove to be more important psychologically than are more general beliefs about the overall level of complexity or simplicity of knowledge. Consider a student reading a microbiology textbook presenting a variety of biological mechanisms. We predict that a student who understands the specific structure of the mechanisms and who adopts the aim of constructing knowledge organized through these mechanisms is likely to use learning strategies appropriate to constructing these mechanisms and judge the resulting knowledge structure according to whether the mechanisms are complete. As a consequence, the student exhibits good performance on learning measures. We further expect that an understanding of the structure of mechanisms in molecular biology will be a better predictor of learning processes and outcomes than will a more general understanding about the complex structure of biological knowledge. A student who appreciates that knowledge is complex but is unfamiliar with the specific structure of the mechanisms of molecular biology lacks the epistemic tools to appropriately organize the material in the textbook. We thus propose that appropriate beliefs about these more specific, fine-grained structures will be more highly predictive of learning than are more generic beliefs about the structure of knowledge.

As another example of these more specific epistemic structures, consider the structure of causation—a form of knowledge that has been intensively examined by philosophers (Cartwright, 2004; Woodward, 2003). Grotzer and Basca (2003) have investigated students' tacit beliefs about the structure of causality in biology. They found that those who understood causal knowledge in a linear way learned less than those with an understanding of causal knowledge as reciprocal and nonlinear. Thus, tacit commitments about causal structure were strong predictors of students' learning.

Collins and Ferguson (1993) called these kinds of knowledge structures *epistemic forms*, which can include structures as diverse as lists, tree structures, and stage models. Hammer and Elby (2002; Rosenberg et al., 2006) have discussed forms of this sort, including lists and explanations. These forms are not situation specific, as a given form may be used across many situations (e.g., a mechanism structure is used widely in many areas of biology; lists may be used in many school settings), but they are more specific than general beliefs about dimensions such as the simplicity and complexity of knowledge. An important aspect of the development of EC may be the development of a repertoire of epistemic forms for understanding the world and engaging in inquiry.

Conclusions

Most EC researchers have investigated the structure of knowledge as a single dimension of belief ranging from simple knowledge to complex knowledge. Our framework includes additional dimensions of belief about the general structure of knowledge. It also includes beliefs about the finer grained knowledge structures such as scientific mechanisms and causal structures. Philosophers have found such specific epistemic structures to be central to the epistemic practices in science, in particular. We believe that EC research would profit by investigating how these more specific epistemic forms affect learning.

Sources, Justification, and Epistemic Stances

The third of the five proposed components of epistemic cognition comprises a cluster of interrelated cognitions related to the sources and justification of knowledge and the stances taken toward knowledge claims (stances such as doubting a claim or believing it to be certain). EC researchers have already extensively investigated cognitions (especially epistemological beliefs) about the sources, justifications, and certainty of knowledge (see Buehl, 2008). However, philosophers discuss many issues that have not yet featured in EC research; we discuss a few of these in this section. A main theme in this section is that epistemologists discuss the sources and justification of knowledge using finer grained, more situation-specific cognitions and that EC research could benefit by investigating these cognitions. In this section, we discuss sources first, followed by justification and then epistemic stances.

Sources

Sources of knowledge refer to the origins of knowledge—such as perception, reasoning, and the testimony of other people. EC researchers who study epistemological beliefs related to sources have investigated constructs including authority or omniscient authority (Hofer, 2000; Kardash & Scholes, 1996; Muis & Franco, 2009; Schommer, 1990), authority/expert knowledge, personal justification for knowing (Hofer, 2000), personal experience or inquiry (McDevitt, Sheehan, Cooney, & Smith, 1994), rationalism (Muis, 2008), and rules of inquiry and the evaluation and integration of multiple knowledge sources (Strømsø, 2009).[7] Muis and Franco (2009) investigated whether knowledge is believed to be sourced outside the self or by active construction within the self. In most studies of beliefs about sources, beliefs are regarded as ranging along a continuum from a pole (such as a belief in personal experience as a source of knowledge) to a pole viewed as its opposite (such as a belief in authority as a source of knowledge).

[7]It appears to us that the constructs investigated by EC researchers as "sources" and those investigated as "justifications" are often the same (e.g., "authority" and "experience" may be treated as a source in one study and as a basis for justification in another). Most of those in this list would be viewed by philosophers as sources. Hence, we have mixed categories of sources and categories of justification together in generating this list.

Philosophers have conceptualized sources in ways that differ significantly from the current practices of EC researchers. Accordingly, the sources component of our framework differs from most current EC conceptualizations in the following respects: (a) It incorporates a broader range of sources that should be investigated. (b) It reconceptualizes the source of authority as testimony. (c) It views sources as interactive. Two sources should not be viewed as opposite ends of a single continuum; rather, multiple sources are simultaneously operative and interact with each other. (d) It recommends examining beliefs about the grounds for trusting sources such as testimony. Further, the ways in which sources interact is situation specific; hence, our framework indicates that EC researchers should examine how learners understand the interaction of sources in different situations. We discuss each of these issues next.

A broader range of sources. Our framework incorporates a broader range of sources than EC researchers have investigated. In the philosophical literature, five sources of knowledge are most widely discussed (Steup, 2005): perception, introspection, memory, reasoning, and testimony (acquiring beliefs from the claims of others). When knowledge has a *perceptual* source, it arises through the faculties of one of the five senses. *Introspection* refers to people's examination of the contents of their own minds and thus produces knowledge of one's own internal experiences. *Memory* is a critical source of knowledge; much knowledge originally acquired in a person's past can be known only with the aid of an accurate memory. For instance, a person's knowledge that she drank a cup of coffee 2 minutes ago depends on an accurate memory of this event. *Reasoning* as a source of knowledge refers to the application of rules (such as logic) to produce knowledge. Finally, knowledge that has *testimony* as a source is learned from others. Philosophically, the term *testimony* is used not in the everyday sense of *testimony at a trial* but in a broader sense; testimony simply refers to all social forms of sharing information and knowledge with others. The developmental psychologist Harris has investigated children's learning from testimony in this broader sense (e.g., Harris & Koenig, 2006).

Although these are the main five sources discussed, philosophers also discuss other sources, including intuition, revelation, sacred scriptures, special mystical or religious experiences, fiction, art, and findings of research (Code, 1991; Plantinga, 2000; Williams, 2002; Zagzebski, 1996). Given the importance of religious beliefs to many students, sources such as revelation and sacred scriptures may be psychologically important sources that are important in determining what students believe about many topics. Epistemologists have pointed out that great literature delivers important truths about topics such as human relationships, what it means to be human, and many other topics (Code, 1991; Zagzebski, 1996). Given the focus of schooling on reading literature, it would be interesting to know whether students find literature to be a source of any significant truths, and if so, which truths they consider might be learned from literature and fiction.

The two sources most commonly investigated in EC research appear to be variations on authority and experience. However, depending on how participants in studies interpret the word *experience*, the term may meld together numerous sources discussed by philosophers—perception, introspection, intuition, reasoning, memory, and even findings of research. EC researchers could enrich their studies by exploring a broader range of sources.

Reconceptualizing the source of authority as testimony. As noted earlier, EC researchers have frequently investigated students' belief in a source of knowledge they refer to as *authority*, referring to authoritative sources such as experts, teachers, or textbook authors. Learning from others seems to be viewed as less desirable than grounding knowledge in one's own experiences or reasoning. Contemporary epistemologists take a sharply different view (Coady, 1992; Lackey & Sosa, 2006), observing that the vast majority of what people know is learned from others; there is simply no feasible way for people to verify through their own experiences most of what they learn from testimonial sources (Coady, 1992; Kusch, 2002). For example, most people's knowledge of local, national, and world affairs depends almost exclusively on the testimony of the media and associated sources. Scientists' knowledge of their own areas of expertise derives largely from the testimony of their colleagues, because they read and hear about many more experiments than they can conduct themselves. Scientists' knowledge even of their *own* experimental results rests crucially on testimony (Lipton, 1988)—the testimony of lab technicians, for example, who attest that data were accurately recorded and that proper experimental procedures were followed. Thus, human knowledge relies pervasively on testimony as a social source of knowledge, even within people's own areas of expertise (Craig, 1990; Kusch, 2002; Williams, 2002). The negative connotation of the term *authority* in much EC research seems to be inconsistent with the important role of testimony in creating and spreading human knowledge.

We recommend, therefore, that EC researchers conceptualize learning from others as using the source of *testimony* rather than the source of *authority.* By recognizing that testimony plays an indispensible role in nearly all human knowledge, EC researchers can explore conditions under which learners' beliefs about learning from testimony facilitate as well as impede learning. Along these lines, Bråten and his colleagues (2011) reported that a belief that knowledge comes from authority (*testimony*, in our terms) is in some circumstances associated with more effective and sophisticated learning strategies when reasoning about multiple documents. Other recent EC work has emphasized that people must rely on experts for knowledge outside their own spheres of experience (Bromme, Kienhues, & Stahl, 2008; Porsch & Bromme, in press; Sinatra & Nadelson, 2011). We would add that experts themselves rely very heavily on social sharing of information even in their own areas of expertise.

Sources interact. In contrast to EC researchers, philosophers emphasize that sources interact. All knowledge derives not from one source versus another but from multiple sources simultaneously. For example, a physics student's knowledge of topics such as forces and friction rests *jointly* on the sources of his or her perceptual experiences, the testimony of teachers and scientists, reasoning about course readings and experiences, and memory needed to retain all these past knowledge-acquisition episodes until the present. We expect that, if given an opportunity to identify multiple sources of particular pieces of knowledge, many students will prove to be well aware of these multiple sources. For example, they may grasp that their knowledge of heat and temperature comes jointly from their teacher, their textbook, their observations, and their reasoning when completing lab reports. Further, they may understand that, in contrast to their knowledge of electricity, which is based partly on many experiments performed in class, their knowledge of General Relativity rests much more on the testimony of scientists, as there are very few school experiences that could empirically demonstrate relativistic effects. Epistemic sophistication lies not in naively thinking that knowledge comes from one source or another but in recognizing how different sources come together to support different knowledge claims (cf. Bromme et al., 2008).

Investigating more fine-grained beliefs about sources and their grounds. We think that EC researchers have frequently investigated beliefs about sources at a grain size that is too coarse to accurately predict or explain students' learning processes and outcomes. The framework we present here indicates the importance of examining in more detail students' grounds for trusting various sources such as testimony.

Consider two hypothetical high school students who score high on a scale measuring belief based on "authority," containing items such as "If my personal experience conflicts with ideas in the biology textbook, the book is probably right" (Hofer, 2000, p. 390). Student A believes what the textbook says simply because she considers textbook authors to be "smart." Student A knows little of the kinds of evidence or methods that scientists use and considers all textbook authors to be equally trustworthy. Student B, in contrast, believes that textbooks reflect biologists' current consensus, supported by a large body of empirical evidence generated through methods that she has some understanding of and believes to be valid. She believes that scientists are well warranted in their beliefs through their extensive, shared empirical investigations. Thus, although agreeing that the textbooks are "probably right," Students A and B differ sharply on their deeper grounds for trusting textbooks.

Although both students score the same on the coarse-grained measure of belief in whether the textbook is probably right, a more fine-grained measure of their beliefs would reveal their very different ideas about the grounds for judging that the textbook is trustworthy. This measure might consist of written or interview follow-up questions asking the students to explain why they think the textbook is probably right. We expect that the two students' different, finer grained grounds for believing textbooks will lead the students to learn and reason very differently.

For example, imagine that these two students read material that conflicts with their textbook. Specifically, they both read a health magazine article making claims about vitamin D that contradicts several of the textbook's assertions. Student A decides that she will simply check the magazine's claim against the textbook. She identifies discrepancies between the two sources and, as a result, summarily dismisses the magazine claim because she considers biology textbook authors to be smarter than journalists. In contrast, Student B begins with an awareness that the study discussed in the magazine is the kind of study that health scientists use to support their conclusions and that this study was conducted after her textbook was published. Hence, she decides to evaluate the methodological quality of the study and concludes that it is credible. As a result, she adopts a stance of uncertainty for the time being. Thus, the two students carry out very different cognitive processes, and their final learning outcomes (their final belief about vitamin D) also differ. To explain these differences, one needs more detailed, fine-grained information about the two students' understanding of textbook testimony as a source of knowledge. Coarse-grained measures fail to capture the very different grounds the students have for believing textbook testimony and so cannot predict these different learning processes.

Some educational researchers have investigated students' ratings of the trustworthiness of testimonial sources (Bråten, Strømsø, & Britt, 2009) and have probed for reasons why students trust these sources (e.g., Mason & Boldrin, 2008; Rouet, Britt, Mason, & Perfetti, 1996). We suggest that researchers further investigate learners' deeper grounds for trusting different sources of knowledge.

Justification

Our EC framework draws on several strands of philosophical research that argue for studying justifications at a finer grain size. EC researchers have investigated epistemic beliefs about types of justifications at a relatively coarse grain size—such as justification by authority, experience, personal justification, rationality, and rules of inquiry (see, e.g., Buehl, 2008). In contrast, our framework specifies that the types of justifications that actually affect people's learning and reasoning are likely to be less general and more specific than these constructs. In this section, we first argue for treating justification at a more specific, finer grain size of analysis. Then, drawing on philosophical work, we discuss justificatory standards—the specific standards people use to evaluate knowledge claims—as one promising avenue of EC research.

Examining justifications at a finer grain size. We contend that beliefs about justification at the broad level of "personal experience" or "rules of inquiry" are at a grain size that it is too coarse to explain important differences in people's epistemic behavior. For example, consider "rules of inquiry" as type of justification (Bråten et al., 2011). Quantitative and qualitative researchers in the social sciences may agree that they are both using "appropriate rules of inquiry," but the rules they use are so very different that they not only conduct completely different forms of inquiry but often come to very different conclusions. To understand the differences in their inquiry practices and conclusions, one must understand the differences in the specific rules of inquiry that are advocated.

As a school-based example of the importance of moving to a finer grain size in examining beliefs about justification, imagine two students who both recognize the importance of evidence in justifications and so would score high on a coarse-grained measure of belief in "justifications based on evidence." These students might nonetheless exhibit very different ideas about what it means to justify ideas based on evidence. When presented with the task of evaluating a nutritional claim about a diet, Student A is committed to the criterion of evaluating claims against as much of the available scientific evidence as possible. This student seeks out many scientific studies and refrains from drawing a conclusion until a broad range of evidence has been explored. Because the scientific evidence points to negative health effects of the diet, Student A concludes tentatively that the diet is likely to be harmful. In contrast, the second student deploys a different evidential criterion—relying principally on personal testimonials as evidence. This second student finds testimonials on several websites, and concludes on this basis that the diet is beneficial. The two students use different learning strategies and end up with different beliefs due to their different evidential standards, even though they score the same on the coarse-grained measure of "belief in justification based on evidence." To understand actual learning processes, one needs measures that probe students' more specific evidential standards, such as what kind of evidence is good evidence in a particular context; interviews might be best suited to access students' finer grained ideas about exactly how evidence can and should be used.

The use of multiple justificatory standards. Philosophers have discussed justificatory principles at a finer grain size than "consistent with rules of inquiry" or "consistent with evidence" (Alston, 2005; T. S. Kuhn, 1977; Newton-Smith, 1981). They have often referred to these as *standards*.

A number of these standards involve different ways in which evidence is used to justify beliefs. For example, one could prefer explanations that explain the greatest range of data (even if there are some anomalous pieces of evidence that cannot be accounted for) or, instead, explanations that explain a narrower range of evidence but do so with few or no anomalies left unaccounted for. One could prefer mathematical precision in predictions, thus privileging this kind of evidence over other kinds of evidence. One could prefer large quantitative studies employing statistical tests as evidence or, instead, detailed case studies. Disagreements in the use of specific justificatory standards related to evidence may help explain differences in how scientists (and others) interpret and evaluate evidence (T. S. Kuhn, 1977).

Philosophers have discussed nonevidential standards for justifying beliefs as well. One such standard is that beliefs are justified to the degree to which they cohere with other established beliefs. Other standards refer to the simplicity of a belief system (e.g., other things being equal, scientists often prefer simpler theories), its internal logical consistency, its "elegance," how understandable it is to other scientists, and its fruitfulness in opening up new lines of research (Laudan, 1984; Newton-Smith, 1981).

There is psychological evidence that at least some of the standards discussed by philosophers are used by students as young as the first grade. Samarapungavan (1992) conducted a study showing that even first graders prefer explanations that conform to standards that philosophers have studied, including scope of evidence covered, consistency with evidence, and simplicity. Pluta, Chinn, and Duncan (2011) asked seventh graders to generate lists of standards for good explanatory models. The students produced a broad range of standards, including many of the evidential and nonevidential standards discussed by philosophers. They tended to favor nonevidential standards such as having an appropriate level of detail, and only about one fourth of the students emphasized evidential standards, suggesting that these standards are not salient to many students.

To understand how students create and evaluate explanations, it will be critical to examine the specific standards that they believe must be met. For instance, science students who favor standards such as "explanations should have an appropriate level of detail" will evaluate explanations very differently from students who favor standards that specify that the explanations must fit all the evidence. Yet all these students may agree, at a coarser grain size of belief, that they are using "reasoning" when evaluating their explanations. To predict and explain how they engage in inquiry in a science class, researchers will need to know their finer grained standards.

Standards are closely connected to the epistemic values that we discussed earlier; in fact, many philosophers use the term *values* to refer to what we are calling justificatory standards (e.g., Laudan, 1984). The individual who uses the standards of "simplicity" to evaluate theories values simplicity in theories. The individual who judges explanations based on whether a mechanism is present values mechanisms. This is another point at which the different components of our network of epistemic cognitions are interconnected.

Situational variation in justificatory standards. A critical point raised by philosophers is that there are no hard-and-fast rules for applying standards to particular situations (e.g., T. S. Kuhn, 1977). Suppose that a student prefers explanations that (a) explain the broadest possible scope of strong evidence (weak evidence may be excluded) and (b) are as simple as possible. To understand how this student uses these two standards to evaluate competing explanations, an EC researcher will need to understand the details of how she applies the standards to particular situations. One set of relevant questions includes how she goes about determining what counts as evidence in particular situations, what range of evidence she decides to consider, how she judges whether the explanation explains the evidence, and how she judges whether the evidence is strong or weak. Another set of questions asks how she evaluates whether a particular explanation is to be regarded as simple, and how "as simple as possible" is to be interpreted in a particular situation. The student's judgments are likely to vary from situation to situation. For example, her contextual criteria for judging whether evidence is strong or weak will differ when she is evaluating forensic evidence in a murder trial as opposed to eyewitness testimony at the same trial, because the kinds of evidence under consideration are different. Similarly, her contextual criteria for evaluating simplicity will differ when she is considering explanations for heat transfer as opposed to explanations for the latest housing bubble. There are no hard-and-fast, situation-free rules for determining what makes evidence strong or weak or what makes an explanation simple. On our framework, people's decisions about what to believe are dependent on how all these contextual issues are worked out. EC researchers can understand how people learn and reason only if they understand the specific justificatory standards that people use and how they apply and interpret these standards in various situations.

Changes in the social context can affect the amount and type of justification needed to support the exact same knowledge claims (DeRose, 2009). For example, the level of support needed to justify the claim that cell phones cause brain tumors is less strict when a person is conversing with friends than when the person is presenting a keynote address at a meeting of the American Medical Association. In addition, the types of justifications that the person offers will likely change across these two situations; anecdotal reports may be accepted as strong evidence in a conversation among friends but not at a medical conference. These contextual variations are another source of variation in justificatory standards. An important task of EC research is to investigate how and why people's commitments regarding adequate justification vary across different contexts.

Epistemic Stances

EC researchers have typically conceptualized "certainty" as a characteristic of knowledge. In the framework presented here, certainty is instead characterized as one of a number of different stances that one can take toward knowledge claims (Chisholm, 1977). In our framework, certainty is a stance taken by an individual toward knowledge claims that are viewed as extremely well justified. Conversely, an individual may adopt a stance of being uncertain about knowledge claims that are considered to be poorly justified. We group "certainty" with "justification" because, for many people, there will be a close relationship between the extent to which they view knowledge claims as certain and the extent to which those claims are judged to have strong justificatory support.

In addition to considering the degree of certainty that a person could take toward knowledge claims, philosophers have identified a variety of other stances that people may take. Laudan (1996) noted that scientists may entertain ideas, consider them, or utilize them as a working hypothesis or assumption. People might adopt additional stances such as withholding judgment or viewing a claim as partly true. People undoubtedly take all these stances at different times toward different knowledge claims. The same person may view as certain the claim that the heart pumps blood, doubt the claim that caffeine harms the heart, and take as a working assumption that vitamin E prevents heart disease. EC research could be amplified by new investigations into the different stances that inquirers take toward claims as well the conditions under which they take different stances.

Conclusions

In summary, in comparison to most current EC research, our framework (a) incorporates a broader array of sources, (b) treats testimony as a potentially valuable source, (c) emphasizes that multiple sources interact to support knowledge, and (d) offers a more detailed account of students' understanding of sources (such as the grounds students have for believing testimony). Our framework also advocates examining specific standards used in justification as well as patterns of contextual variation in how these standards are applied in various situations. Making finer grained distinctions in models of EC would provide researchers with a better basis for predicting and explaining variance in learning processes and outcomes.

Epistemic Virtues and Vices

The fourth proposed component of EC comprises a set of interrelated cognitions related to *epistemic virtues and vices.* In this section, we explain this component and show how our framework goes beyond current EC research on this topic.

Virtue epistemology is a flourishing area of research in contemporary philosophy, with roots in the writings of Plato, Aristotle, Aquinas, and Dewey (Zagzebski, 1996). Reviewing the many varieties of virtue epistemology is beyond the scope of this article, so we build on an approach developed by Linda Zagzebski (1996, 2009) and others (e.g., Montmarquet, 1986), who ground their theories of epistemic virtues in virtue theories of ethics.[8] On these theories, the achievement of knowledge and understanding flows from epistemic virtues such as intellectual honesty and intellectual courage. An *epistemic virtue* is a learned, stable disposition that is (a) directed at epistemic aims such as true belief, knowledge, and understanding and (b) relatively efficacious in achieving these aims. An example of an epistemic virtue is *intellectual carefulness* (Zagzegski, 1996). An inquirer disposed to be intellectually careful will painstakingly record information gathered in inquiry, making sure that no errors are made and that nothing is overlooked. Intellectual carefulness is an epistemic virtue because a willingness to take care in gathering information and reaching conclusions demonstrates a commitment to the epistemic aim of achieving true beliefs; those who exemplify this virtue are more likely to develop true beliefs than are those who are perfunctory and prone to error when gathering evidence. Epistemic virtues are thus truth conductive or knowledge conducive.

Virtue epistemologists discuss epistemic vices as well as virtues (Zagzebski, 1996). An *epistemic vice* is the opposite of a virtue; epistemic vices impede rather than facilitate the attainment of knowledge and understanding. Examples include intellectual cowardice and closed-mindedness. Whether a disposition can be regarded as a virtue or a vice hinges on its efficacy in effecting the achievement of epistemic aims. Open-mindedness is generally considered an epistemic virtue because it enables people to set aside false beliefs and instead adopt better-supported views. Intellectual sloth is generally considered an epistemic vice because a person who makes little effort to gain knowledge is unlikely to achieve many true beliefs. However, as we discuss later, it should be noted that judgments of whether dispositions such as open-mindedness should be regarded as an epistemic virtue or vice can vary according to the context. A person who keeps an open mind about whether humans are really animals is flouting the evidence to the extent that her open-mindedness is a vice rather than a virtue.

A consequence of treating dispositions such as open-mindedness as epistemic virtues, akin to ethical virtues, is that we can readily account for the language of praise and blame that is frequently used to describe people's epistemic

[8]There are two broad, complementary approaches to virtue epistemology (Alston, 2005; Greco, 2009). We discuss one approach in this section. The other approach views epistemic virtues as aspects of cognitive faculties such as perception, memory, and reasoning; intellectually virtuous people are those with well-functioning cognitive faculties such as good perception, good memory, and good reasoning. It is thus widely considered to be a form of reliabilism, which we discuss in the next section.

character. People praise a person for having the intellectual courage to stand up for an unpopular but correct belief because they are regarded as exercising a virtue of character. Conversely, people may choose to blame a person who exhibits intellectual cowardice, which is considered to be an ethical weakness of character. The ubiquity of such evaluative terms in everyday language suggests that epistemic virtues are an integral feature of people's everyday thinking about epistemic matters.

Several EC researchers have begun studying constructs that are similar to epistemic virtues and vices, which they have referred to as "dispositions" (Sinatra & Kardash, 2004) or thinking dispositions (N.-M. Chan, Ho, & Ku, 2011; Sinatra & Chinn, in press; Stanovich, 1999). Many of the thinking dispositions studied are, according to our framework, epistemic virtues (including dispositions to open-mindedness and flexible thinking) or epistemic vices (including dogmatism). Researchers have found significant correlations between these measures and measures of performance on a variety of academic tasks (Nussbaum & Sinatra, 2003; Sinatra, Southerland, McConaughy, & Demastes, 2003). In one study, performance on an argument evaluation task was positively associated with scores on measures of open-mindedness (an epistemic virtue) and negatively associated with measures of dogmatism (an epistemic vice; Stanovich & West, 1997). Kruglanski and colleagues (e.g., Kruglanski & Webster, 1996) have extensively investigated the need for closure, a disposition that is akin to an epistemic vice (see DeBacker & Crowson, 2009). Individuals who score high on need for closure desire definite answers to questions and are averse toward ambiguity (Kruglanski & Webster, 1996). Measures of this construct predict performance on many tasks. For example, people who differ in their need for closure differ in how readily they change their minds under various circumstances (Kruglanski & Webster, 1996).

To date, only a few EC studies have incorporated measures of epistemic virtues or dispositions. In contrast, virtue epistemologists view epistemic virtues as central to the tapestry of topics investigated in epistemology. On this view, understanding how people achieve and fail to achieve their epistemic aims requires understanding their particular array of epistemic virtues and vices as well as the ways in which these help shape belief formation, understanding, inquiry, and action. Thus, our framework places a much stronger emphasis on investigating the role of epistemic virtues and vices within epistemic cognition.

The virtues component of our framework includes elements that go beyond current EC research. First, following contemporary philosophers, we include in our framework many more virtues and vices than EC researchers have as yet investigated. Second, our framework is more specific about what counts as an epistemic disposition, as opposed to a general thinking disposition. Third, our framework posits that beliefs about virtues are relevant to EC research, in addition to the virtues themselves. Fourth, we note the importance of studying the context specificity of virtues and vices. We discuss these extensions next.

Expanding the Range of Virtues and Vices Investigated

EC researchers have focused on several epistemic virtues and vices: the virtues of open-mindedness (N.-M. Chan et al., 2011; Sinatra & Kardash, 2004), conscientiousness (N.-M. Chan et al., 2011) and the vices of dogmatism (Sinatra & Kardash, 2004), unwillingness to give up beliefs (Sinatra & Kardash, 2004; Stanovich & West, 1997), and the need for closure (DeBacker & Crowson, 2009). Virtue epistemologists have examined a broader range of virtues and vices that could also be investigated by EC researchers. Virtues discussed by Zagzebski (1996) include "intellectual carefulness, perseverance, humility, vigor, flexibility, courage, and thoroughness, as well as open-mindedness, fair-mindedness, insightfulness, and the virtues opposed to wishful thinking, obtuseness, and conformity" (p. 152). Montmarquet (1986) postulated three core epistemic virtues: impartiality (a willingness to exchange ideas with others and to learn from them, open-mindedly, without jealousy or bias, and with a lively sense of one's fallibility), intellectual sobriety (being a careful inquirer who accepts only what is warranted by the evidence), and intellectual courage (examining alternatives to popularly held beliefs; perseverance in the face of opposition from others until one is convinced one is mistaken). Williams (2002) posited two core epistemic virtues—sincerity and accuracy—and argued that these two social virtues are vital to the good functioning of a community, whose members will benefit from sharing knowledge with each other.

Expanding the range of virtues and vices investigated will provide EC researchers with the potential to better predict and explain learning processes. Some of these virtues and vices may prove to be better predictors of learning processes in particular situations than those that have already been studied. For example, students who have the disposition to exemplify the epistemic virtue of intellectual courage may be more likely than other students to engage in the strategy of reasoned argumentation in groups, particularly when they are advocating a minority position. As another example, students who express a strong commitment to the epistemic virtue of intellectual carefulness can be expected to set goals to check information carefully, to engage appropriate strategies, and then to apply standards to check their results. This virtue may be highly predictive of learning processes and knowledge gains in a science laboratory class involving the collection of many data points. In sum, by exploring a wider range of virtues and vices, EC researchers can provide important information about the ways in which learners' characters and dispositions can serve to aid or undermine their attainment of valuable epistemic aims.

Distinguishing between epistemic virtues and vices and other dispositions. Not all scales that assess cognitive dispositions are measures of epistemic virtues. Our framework provides conceptual resources for distinguishing epistemic virtues and vices from other dispositions. An example is the Need for Cognition scale (Cacioppo & Petty, 1982), which has been used by several educational researchers (e.g., Nussbaum, 2005). The Need for Cognition scale measures the degree to which individuals engage in effortful cognitive activity but includes a number of items that are not properly epistemic, such as the item "I find satisfaction in deliberating hard and for long hours." A person who agrees with this item clearly enjoys thinking, but there is no indication that the hours of deliberation are intended to promote epistemic aims such as truth or understanding; the individual might enjoy spending hours solving difficult crossword puzzles for fun. To be an epistemic virtue, however, a disposition must be directed at an epistemic aim.

Another partial example is the need for closure construct, which is composed of five subscales: Preference for Order, Predictability, Decisiveness, Discomfort With Ambiguity, and Closed-Mindedness. Although we agree that Closed-Mindedness and Discomfort With Ambiguity are clear-cut epistemic vices, the other three are far less clearly so. To be a vice, it must be clear that the disposition ordinarily impedes the attainment of an epistemic aim. But it is unclear to us, for example, that a preference for order necessarily impedes successful inquiry. One can imagine that a scientific team that maintains an orderly laboratory may be more likely to make new discoveries than a team with a chaotic lab.

Our analysis of these two scales highlights an important feature of our framework: Cognitions are epistemic only if they are directed at epistemic aims or accomplishments. Although we are proposing an expansion of the topics considered to be part of EC, it is a limited expansion that avoids the danger of treating *all* cognitions as epistemic. However, we do not want to suggest that EC researchers should refrain from studying interactions of epistemic and nonepistemic dispositions. Nonepistemic dispositions such as Need for Cognition may interact with epistemic virtues and epistemic beliefs to influence students' learning and reasoning processes, and researchers should continue to investigate a broad array of constructs.

Beliefs About Virtues and Vices

To date, EC researchers have investigated students' epistemic virtues and vices by employing self-report measures of students' dispositions to act in epistemically virtuous or vicious ways. Philosophical investigations suggest another approach that has not been used: evaluating students' reflective beliefs regarding epistemic virtues and vices (Morton, 2006). Researchers might present students with vignettes so as to probe their judgments about people who display epistemic virtues. For example, to investigate a virtue such as intellectual courage, researchers could construct vignettes in which a character (perhaps a student) must decide whether to undertake an act of intellectual courage, with students indicating what the character should do as well as explaining why.

Students' ideas about epistemic virtues and vices might vary greatly and might also diverge from expert views. Zagzebski (1996) argued, for example, that believing on faith without evidence is an intellectual vice, but some students might view such dispositions as epistemically virtuous. Similarly, vignette studies might indicate that students vary greatly in whether they view open-mindedness as a virtue, with some instead regarding sticking doggedly to a view as epistemically virtuous. Such students are likely to reason very differently from those who regard open-mindedness as a virtue. Students who consider open-mindedness to be a virtue but fail to exemplify open-mindedness in classroom tasks may be receptive to encouragement to be more open-minded. In contrast, students who believe open-mindedness to be a vice are unlikely to become more open-minded just because a teacher encourages them to do so. They may instead first need to be persuaded that open-mindedness is a productive means for the achievement of their epistemic aims.

Context Specificity of Epistemic Virtues and Vices

To this point, we have treated epistemic virtues and vices as relatively stable motivational dispositions. But epistemologists note two respects in which epistemic virtues and vices are contextual, varying from situation to situation. First, like psychologists who distinguish between states and traits, epistemologists observe that even though a person may tend to exemplify a particular virtue, they will often fail to do so; whether they exemplify a particular virtue depends partly on the situation (Zagzebski, 1996). For example, a person may be open-minded when there is enough time to reflect but not when time is pressing (cf. DeBacker & Crowson, 2009). A person may be open-minded in conversations with friends but not with colleagues, or about some scientific topics but not about her beliefs about herself. Del Carlo and Bodner (2004) found that chemistry students view fudging data on lab reports to be cheating (a vice, in our terms) if scientists do it but not if they do it themselves in their own lab classes. Social psychologists have found great situational variation in many personality traits and dispositions (Mischel, 2004). We expect that this will be true of epistemic virtues and vices as well. A vital task for EC research is thus to investigate situational variation in people's exemplification of epistemic virtues and vices.

Second, virtue epistemologists have observed that whether a particular disposition can even be viewed as virtuous can depend on the context (Zagzebski, 1996). If health activists persevere against popular opinion and eventually persuade the scientific community and the public that a particular food additive is harmful, they are likely to be viewed as exemplifying epistemic virtues including perseverance and intellectual courage. If, on the other hand, flat-earth activists persevere against popular opinion to promote their views, the same kinds of behaviors are likely to be viewed instead as exemplifying epistemic vices such as closed-mindedness. Analogous behaviors can thus be evaluated entirely differently, depending on the context.

We believe it would be productive for EC researchers to examine learners' ideas about when different behaviors are considered to be epistemic virtues or vices and when they are not. For example, by understanding when learners think it is appropriate to be open-minded and when it is not, researchers may be able to better predict when learners will engage in open-minded learning processes and when they will not. Similarly, by understanding when learners think that it is appropriate to be intellectually careful and when there is no need to be careful, researchers can better predict when they will use the strategies and evaluative standards associated with intellectual care. To our knowledge, there has been no research on these types of issues.

Conclusions

Epistemologists have presented a strong philosophical case for considering epistemic virtues and vices as a component of epistemic cognition. There is a small but growing body of psychological research on several important epistemic virtues and vices. As an implication of our framework, we recommend that these investigations be intensified and expanded to investigate a wider range of virtues and vices. We also recommend that scholars investigate the context specificity of students' epistemic virtues and vices as well as their beliefs about their situational variation. Little EC research has as yet addressed either form of contextual variation.

Reliable and Unreliable Processes of Achieving Epistemic Aims

The fifth component of epistemic cognition comprises a cluster of interrelated cognitions related to *reliable and unreliable processes of achieving epistemic aims*. As an example of such cognitions, consider a college student, Melissa, who has a set of beliefs about reliable and unreliable processes of achieving true beliefs. Melissa believes that normal human reasoning is subject to biases such as confirmation bias and thus believes that typical uncorrected human reasoning is an *unreliable process* for achieving knowledge. In contrast, Melissa believes that a *reliable process* for achieving knowledge involves seeking out a broad range of evidence and paying special heed to data that contradicts her preferred theories. Melissa's beliefs are about the processes (such as the reasoning strategies) for achieving her epistemic aims. These beliefs can be viewed as beliefs about *processes of knowing*, one of the two basic categories in the EC model advanced by Hofer and Pintrich (1997). The cognitions in our fifth component of EC focus on the causal processes—including strategies and other procedures and activities—by which one can achieve knowledge, understanding, and other epistemic aims.

In this section, we begin by briefly discussing the philosophical theory of reliabilism, which provides the conceptual basis for the fifth component of our framework. We then discuss implications for EC research.

Reliabilism

The philosophical warrant for including cognitions about reliable and unreliable processes of knowing as part of epistemic cognition derives from an epistemological movement known as *reliabilism* (Bishop & Trout, 2005; Dretske, 2000; Goldman, 1976, 1986, 1999; Kornblith, 2002; Sosa, 2001). Reliabilists focus on the processes by which true and false beliefs are produced. For reliabilists, a belief is justified (roughly speaking) if it results from a reliable belief-forming process or set of processes (Bishop & Trout, 2005; Goldman, 1976, 1999). As an example of a true belief produced by a reliable process, consider Maria, who is picnicking in a sunny meadow and forms the belief that the grass on which she sits is green. For a reliabilist, Maria is justified in her belief because her visual and cognitive processes are a reliable guide to the color of the grass in her current environment. That is, under normal lighting conditions, her visual and cognitive processes are very likely to produce true perceptual beliefs. On reliabilist theories, Maria's belief is not justified by virtue of any conscious justification that she might be able to offer for it; indeed, her belief is justified even if she can offer no such justification, or only a very inadequate one. Maria's belief is instead justified by virtue of the causal processes (operating through her visual and cognitive systems under appropriate, well-lighted environmental conditions) that reliably produce true beliefs.

One prominent area of research on reliable processes of belief formation examines specific practices of scientific inquiry (Goldman, 1999). By studying the inquiry processes that are implicated in the production of scientific knowledge, philosophers learn about the relative effectiveness of different practices of inquiry. For example, philosophers have examined practices of experimentation (Giere, 1988; Staley, 2004), practices for generating creative new ideas (Nersessian, 2008), and practices used by collaborative teams to coordinate their work effectively (Staley, 2004).

Reliabilism thus brings individual and social practices of discovery and inquiry squarely into the epistemological fold. Epistemologists study "how knowledge ... is produced or generated" (Longino, 2002, p. 78), and so the philosophical study of inquiry is one of their central concerns (Alston, 2005; Hookway, 2006). Goldman (1999) wrote that a central question for philosophers is, "Which practices have a comparatively favorable impact on knowledge as contrasted with error and ignorance?" (p. 5).

Reliabilism is also relevant to justification. Many epistemologists—whether reliabilist or not—would agree that in reflecting on whether a belief is justified, it is valuable to consider the reliability of the processes that generated the belief (Alston, 2005; Haack, 1993; Plantinga, 1993; Sosa & BonJour, 2003). For example, in evaluating whether the conclusions presented in a meta-analysis of research on dieting are justified, it is relevant to reflect on the series of procedures that the analyst used to select articles and conduct the analysis and to evaluate whether these procedures are likely to achieve true conclusions.

Reliabilism is also related to epistemic virtues; as we discussed earlier, epistemic virtues are virtues because they are reliably conducive to achieving epistemic aims. Thus, epistemic virtues can be viewed as an important element of some of the reliable processes by which epistemic aims are achieved.

In the following sections, we discuss implications of reliabilism for EC research and thereby elaborate on the fifth component of our framework. We discuss three main differences between our approach and approaches used in most current EC research. First, we argue that current EC research pays little attention to beliefs about causal processes of belief formation, and we recommend investigation of such beliefs as a central focus of future work in the field. Second, we propose research at a finer grain size, investigating more specific beliefs about the conditions under which causal processes are reliable or unreliable. Third, we suggest a range of new topics suggested by reliabilism for EC research to undertake.

Investigating Learners' Beliefs About Reliable and Unreliable Processes for Achieving Epistemic Aims

A straightforward implication of the philosophical theory of reliabilism is that EC researchers should examine learners' beliefs about reliable and unreliable processes of achieving epistemic aims such as knowledge. To date, despite Hofer and Pintrich's (1997) emphasis on processes of knowing, it appears to us that there has been little research on beliefs (explicit or tacit) about the detailed causal processes by which knowledge is produced. We recommend two general types of research in this area. The first type addresses learners' beliefs about the reliability of the inquiry processes that they themselves can use when seeking knowledge. For example, researchers could examine history students' beliefs about the group inquiry processes that are effective (and those that are ineffective) at promoting the development of good historical explanations on the basis of source documents. The second type addresses learners' beliefs about the reliability of the inquiry processes used by others (such as experts, peers, etc.). For example, although most people do not have the resources, opportunity, or desire to conduct medical research to investigate important health questions, they may nonetheless have beliefs about the reliable inquiry processes that medical researchers should use to conduct this research, and they then can use these beliefs to help them evaluate medical research when they encounter it.

There is some EC research indicating that students can articulate beliefs about the reliable processes by which epistemic aims can be achieved. Baxter Magolda (1992) described some college students' views that diverse perspectives presented in discussion are facilitative of knowledge development; this appears to be a belief about a reliable process of social, peer-stimulated knowledge formation. In a qualitative study, Hofer (2004) described a student who expressed the belief that textbooks are credible because they "have normally been reviewed by many people or written by many people, so you know it's the opinion of more than one person" (p. 153). This is a belief describing a process of textbook development as trustworthy because multiple people are involved. Research by White, Shimoda, and Frederiksen (1999) indicates that, with instruction, students can develop more sophisticated beliefs about social processes for achieving knowledge.

At least two scales have been developed with items focused predominantly on reliable processes of knowledge formation (Bråten & Strømsø, 2010; Kimmel & Volet, 2010). As one example, Kimmel and Volet (2010) employed a scale measuring beliefs about the cognitive benefits of group work, including items such as "Interacting with peers for this group assignment will enrich my knowledge and understanding," which measures belief in peer interaction as a reliable means of achieving two specific epistemic aims. In their study, business students rated peer interaction as less effective at promoting knowledge after experience with group work, suggesting that people modify their beliefs about reliable processes based on their experience. The early development of scales assessing beliefs about reliable processes of knowledge formation indicates the viability of expanded work in this area.

However, questionnaire items such as these do not probe very deeply into beliefs about reliable processes of belief formation. A student may have more detailed beliefs about how and why interacting with peers enriches knowledge that are not tapped by a simple questionnaire item. These more detailed beliefs could be assessed in interviews or inferred from observation of classroom interactions. If a student stated during group work in a science inquiry lesson, "We'd better make sure we discuss both sides if we want to come up with a good explanation," the student has provided evidence of at least a tacit belief that discussing both sides of a question is a good way to develop good explanations.

Investigating beliefs about conditions under which processes of achieving epistemic aims are reliable. Among the few studies that have measured students' beliefs about reliable processes of achieving epistemic aims, all have assessed beliefs at a coarse grain size. For instance, Bråten and Strømsø (2010) developed a scale with items such as "When I read about issues related to climate, I have most trust in claims that are based on scientific investigations." This item assesses a very general belief in the reliability of scientific investigations as a means of producing trustworthy claims to knowledge. In contrast, at a finer grain size, researchers could seek to measure finer grained beliefs about scientific investigations, such as beliefs about whether computer simulation models (often used in climate research) are a reliable means of developing scientific knowledge. Or, using interview protocols, researchers could probe beliefs about the features of scientific investigation that are considered to be reliable; learners might point to more specific inquiry practices such as peer critique and review, careful collection of data, or replication. At a still finer grain size, researchers could investigate not only whether (for example) learners regard peer review as a reliable inquiry process but, more specifically, the conditions under which peer review is thought to either promote or hinder the attainment of knowledge. For example, a person might believe that peer review promotes scientific knowledge, but only under certain conditions: if the review process is blind, if reviewers include representatives of different perspectives, and if the submitted articles include enough detail to permit detection of methodological errors.

Philosophical scholarship strongly emphasizes the importance of examining the specific conditions under which processes for producing knowledge do and do not reliably produce true beliefs (Dretske, 2000; Goldman, 1986, 1999, 2002; Plantinga, 1993). For example, visual perception is often reliable, but not under conditions of dim light or long distances that surpass the perceiver's visual acuity (Dretske, 2000). Similarly, the social practice of deliberative argumentation may be a productive way to advance knowledge, but only under the condition that participants communicate an underlying respect for each other (Goldman, 1999). If some people feel that they are not respected, they may fail to participate, and the group loses input that could help it reach a better decision. To understand processes for belief-formation involving argumentation, one must understand the typical conditions under which social argumentation typically succeeds and under which it fails. More generally, the crucial point is that few, if any, processes are reliable under all conditions. Processes of perception, testimony, and argumentation are all sometimes reliable and sometimes not, depending on the conditions prevailing in a particular situation. We expect that most people are well aware of this and have explicit or tacit beliefs about the various conditions under which different processes of belief formation are reliable and unreliable. These beliefs may, of course, be mistaken, and they are likely to differ from learner to learner.

From a psychological perspective, examining people's beliefs about the specific conditions that make a process reliable or unreliable should allow EC researchers to better predict and explain learning processes as learners engage in both inquiry and evaluation tasks. We illustrate this point with an example of two students' beliefs regarding the reliability of argumentation in promoting knowledge. Both students agree, at a coarse grain size, that argumentation promotes the development of knowledge. However, the two students have different, perhaps tacit beliefs about the conditions under which argumentation is effective. Student A believes that argumentation is effective as long as it is not a "fight." In contrast, Student B believes that for argumentation to be effective, everyone in the group must share ideas, listen to each other, give many reasons, and consider alternative theories and evidence. We suggest that predicting and explaining these students' learning processes and outcomes on particular tasks is likely to require knowledge of each student's beliefs about the conditions under which argumentation is effective.

To illustrate, consider a group inquiry task in which these two students each adopt the aim of achieving knowledge while working in different groups. Consistent with her views about when argumentation is effective, Student A encourages her peers to avoid personal disputes, but she does not encourage them to consider more reasons or alternative theories, as she does not realize that these conditions are necessary for a group to effectively produce knowledge. Other students also fail to encourage deeper reflection, and the group's discussion is shallow. In contrast, and in line with her views about the conditions under which argumentation produces knowledge, Student B encourages her groupmates to talk, and she asks them for their reasons. She advances alternative theories herself as well as encouraging her groupmates to do so, and she draws the group's attention to evidence they had been ignoring. As a result of her encouragement, Student B's group engages in much more extensive discussion of alternative viewpoints, reasons, and evidence than Student A's group, and so it develops a superior group product. To explain how these students engage in this inquiry task, it is necessary to know not only their beliefs about argumentation in general but also the finer grained beliefs about the specific conditions under which they believe argumentation is effective and ineffective. Despite sharing the same belief in the general efficacy of argumentation, the students nonetheless engage in very different interactive processes as a result of their very different beliefs about the conditions that must be in place for argumentation to be effective.

Types of processes of knowledge formation. In this final section of our discussion of reliable processes of belief formation, we briefly discuss a range of the processes that philosophers have discussed. EC researchers could investigate students' beliefs and commitments related to any of these processes. The processes fall into four broad categories: cognitive processes, formal processes for conducting inquiry, interpersonal processes, and community processes. We briefly discuss several of these that could be productive targets for EC research.

Cognitive processes include a variety of the processes that operate within individual cognition. We have discussed perception (the example of Maria earlier in this section) and beliefs about reasoning (the example of Melissa at the very beginning of this section). Other cognitive processes that have been considered by philosophers include those involved in memory, in the generation of new ideas, and in the ways in which emotions enter into belief formation.

As an example of how EC researchers could investigate epistemic cognition involving these processes, we consider emotion and reasoning. A traditional view has been that good reasoning should be dispassionate and free of the contaminating emotions that lead to biases. However, in recent work, philosophers have observed that certain emotions seem critical to knowledge production—emotions such as curiosity and a passion for finding things out (Code, 1991; Zagzebski, 1996). Furthermore, seemingly negative emotions such as anger over being criticized might spur those who are criticized to redouble their efforts to gather evidence, which serves to support knowledge development. People's beliefs regarding whether emotions can be involved in reliable processes of reasoning may be important in understanding how they reason themselves as well as how they evaluate the reasoning of others. For example, people who believe that emotion renders scientific reasoning unreliable may become critical of climate science if they observe climate scientists express anger over distortions of their work in a television interview.

Processes of formal inquiry include processes such as experimentation and other methods of investigation, such as formal opinion polls and surveys, correlational studies, case studies, and so on. Investigations that employ each of these methods feature prominently in the media, in students' textbooks, and in the workplace of many people. To understand how people understand and interpret the results of these investigations, it will be necessary to understand people's beliefs about the conditions under which these types of investigations reliably produce knowledge and understanding as well as their beliefs about the conditions under which they do not. For example, an individual who is aware of how wording changes can affect poll results may tend to be far more skeptical of news reports and more attentive to the details of the polls than someone who is unaware of these conditions.

Prominent among *interpersonal processes* that have been investigated by philosophers is argumentation, which we have already discussed. Philosophers also examine other aspects of small-group procedures, such as the best procedures for making decisions (e.g., voting or seeking consensus), how groups should be structured (e.g., with more or less stringent rules for participation), how much cognitive diversity a group should have to be effective (ranging from all group members having very different initial perspectives to all members starting with the same perspective; Goldman, 1999). People's beliefs about such procedures may influence how they organize group work as well as how they evaluate the products of group deliberations.

Community and institutional processes include processes for funding research, processes of community critique (such as peer review), and processes of sharing knowledge (such as online wikis and the mass media). Given the preeminent role of the media in the creation and dissemination of information, it seems valuable for EC researchers to investigate students' views about how the media operates, how the media ought to operate, and how truth-conducive various media practices are. Philosophers have addressed issues (Cox & Goldman, 1994; Goldman, 1999) such as: How does corporate ownership of media affect what gets reported and how it is reported? What procedures for verifying sources should be used to increase the significant true beliefs available to society? We expect that people's beliefs about reliable and unreliable practices of the media will play a central role in their evaluation of the many claims that the media delivers to them.

Conclusions

There has as yet been very little EC research on people's beliefs about the causal processes by which humans achieve epistemic aims. We argue that EC research could be enhanced through attention to beliefs about the reliable processes by which epistemic aims, like knowledge, are achieved. Researchers could address beliefs about a variety of processes in these four categories: cognitive, formal inquiry, interpersonal, and institutional. People's beliefs about processes for achieving epistemic aims are likely to influence both their inquiry processes and their evaluation of the knowledge claims made by others.

We recommend further that EC research focus on participants' fine-grained beliefs about the conditions under which processes are reliable or unreliable. The few extant measures of beliefs about reliable processes of knowledge formation assess beliefs only in very general processes and do not investigate the specific conditions under which these general processes are considered to be reliable or unreliable. We argue that it is only by attending to specific beliefs about these conditions that EC researchers will be able to understand how people engage in particular learning tasks.

DISCUSSION

In this article, we have presented philosophical and psychological grounds for including five components in a framework for epistemic cognition, with each component comprising a cluster of interrelated cognitions. The components are (a) epistemic aims and epistemic value; (b) the structure of knowledge and other epistemic achievements; (c) the sources and justification of knowledge and other epistemic achievements, and the related epistemic stances; (d) epistemic virtues and vices; and (e) reliable and unreliable processes for achieving epistemic aims. This integrated framework draws extensively on work in contemporary epistemology on the nature and character of epistemic cognition.

In describing each component, we have presented a summary of how our proposed framework differs from much current EC research. In this concluding section, we briefly highlight several implications of our analysis for future research.

Fine-Grained, Situated Epistemic Cognition

Our framework for EC treats epistemic cognition at a finer grain size and with a greater attention to the contexts within which it is embedded. In some instances, the finer grain size derives from a focus on more specific cognitions, as with the focus on specific justificatory standards (e.g., a commitment to specific rules of quantitative and experimental inquiry) instead of more general standards (e.g., a generic commitment to "rules of inquiry"). In other instances, the finer grain size derives from a focus on explicit or tacit beliefs about the conditions under which cognitions are applicable. For example, our framework posits that people have beliefs regarding the specific conditions under which different processes for producing knowledge (e.g., argumentation, perception) are reliable. Our framework also indicates that epistemic cognitions often vary from situation to situation. For example, we argue that contextual judgments are needed to decide how to tailor and apply justificatory standards such as "good explanations are as simple as possible" within particular situations.

The framework we have presented bears some similarities to the context-specific approaches advocated by Hammer, Elby, and their colleagues (e.g., Elby & Hammer, 2001; Hammer & Elby, 2002), who have documented many instances of situational variation in epistemic cognition. We believe that our framework advances work on the situation specificity of epistemic cognition by identifying a variety of specific ways in which it is likely to vary from situation to situation.

In light of our arguments that EC is often situation specific, we believe that an important task for future EC research is to examine patterns of situational variation and to develop explanations for why these patterns exist. Our framework provides theoretical resources for explaining situational variation in EC. If a student applies one set of justificatory standards in one situation and another set in a different situation, the difference may arise from the students' (perhaps tacit) beliefs about the conditions under which each set of standards are applicable. The task of research is then to understand these applicability conditions. In short, we suggest that EC researchers should seek both to understand how epistemic cognition varies across situations and to develop explanations for why this variation occurs.

Methods of Investigation

In recent years, many psychologists have investigated epistemic cognition using questionnaires that ask participants to rate their agreement with domain-general or domain-specific statements (see Buehl, 2008). One methodological implication of our analysis is that finer grained, situation-specific measures of EC components are likely to enhance the prediction of learning and reasoning in particular situations. We have argued that many existing EC measures measure cognitions at too coarse a grain size to explain the variance in performance on learning tasks. However, it may be difficult to capture the nuances of contextualized cognitions in questionnaire measures. Interview and mixed methods may be better suited to understand how and why people's epistemic commitments vary across a variety of situations. Interview measures have the additional advantage of allowing interviewers to probe more deeply into the complex and subtle interactions among various epistemic and nonepistemic cognitions. In addition, we recommend that EC researchers intensify their investigations of tacit epistemic commitments by investigating how people engage in epistemic tasks such as justifying claims, evaluating sources, and the like (e.g., Mason & Boldrin, 2008). Combining different research methods is likely to be a productive way to gain a fuller picture of the complexities of epistemic cognition across different situations.

Because EC encompasses many interrelated cognitions on a wide range of topics, it will likely be impossible to construct any single measure or interview protocol that captures all aspects of EC simultaneously. An overall theory of epistemic cognition will emerge only by piecing together results from many studies, each of which examines a subset of relevant cognitions. This is a consequence of the complex, multifaceted nature of epistemic cognition itself.

Developmental Trajectories

Another application of our framework lies in its use to explore developmental changes in epistemic cognition along the five components we have discussed. For example, researchers could investigate changes in learners' cognitions related to epistemic virtues. They could also investigate growth in learners' awareness of individual and social processes for reliably producing knowledge, as well as changes in beliefs about the conditions under which different processes are reliable. Researchers might also study the interrelations between various cognitions as learners grow older and gain in experience. Our framework thus provides new theoretical resources for researchers aiming to understand developmental change.

Social Aspects of Epistemic Cognition

A very promising area for future EC research is to explore social aspects of epistemic cognition, several of which we have considered in this article. We have discussed beliefs about testimony, a pervasive social source of knowledge. We have also discussed beliefs about reliable social processes (e.g., argumentation, peer review, media processes) for achieving epistemic aims. Future research could also examine cognitions related to social features of the other components of the framework.

A limitation of our discussion of social aspects of EC is that we have focused strictly on *individuals'* cognitions related to the five components of our framework. However, EC researchers could also investigate epistemic cognition at the level of groups. For example, in addition to examining individuals' beliefs about testimony, EC researchers could examine the actual dynamics of how testimony disseminates knowledge among students engaged in inquiry in constructivist learning environments. This would involve looking at epistemic practices of groups of learners, which includes studying how individual practices aggregate, as well as whether the epistemic practices and beliefs of individuals vary depending on social factors.

Interdisciplinary Engagement

A central theme underlying this article is that cross-disciplinary interchange can lead to improved conceptual analyses and a fruitful expansion of the EC research agenda. We have argued that EC research will profit from greater engagement with philosophical literatures. We believe further that vigorous psychological research on the various topics we have discussed will yield findings of which many philosophers will want to take account. Richer connections between the two fields are likely to benefit both.

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